1 Development Suite MES-Cockpit-Services

MES Development Suite offers functions for customizing the MES-Cockpit 3.1 according to the customer's requirements. This document describes the functions provided by the customizing tool MES-Cockpit (MC-DSCS).

Default *.qvw files and scripts must not be changed. Unless otherwise specified, changes must always be made in customer-specific files.

All modifications described here can only be made if the user has purchased the customizing tool (MC-DSCS). It is recommended to complete the relevant customizing training course (CUT-MSC).

1.1 Advanced Analysis Options

Overview of benefits

By default, there are already pre-defined evaluations for the Performance Analysis and Production Monitoring that can be applied by users. These provided options may be customized and modified.

MES-Cockpit uses QlikView to visualize data in the Performance Analysis and Performance Monitoring.

Requirements

The original QlikView document can be found in the following path on the respective MES-Cockpit server:

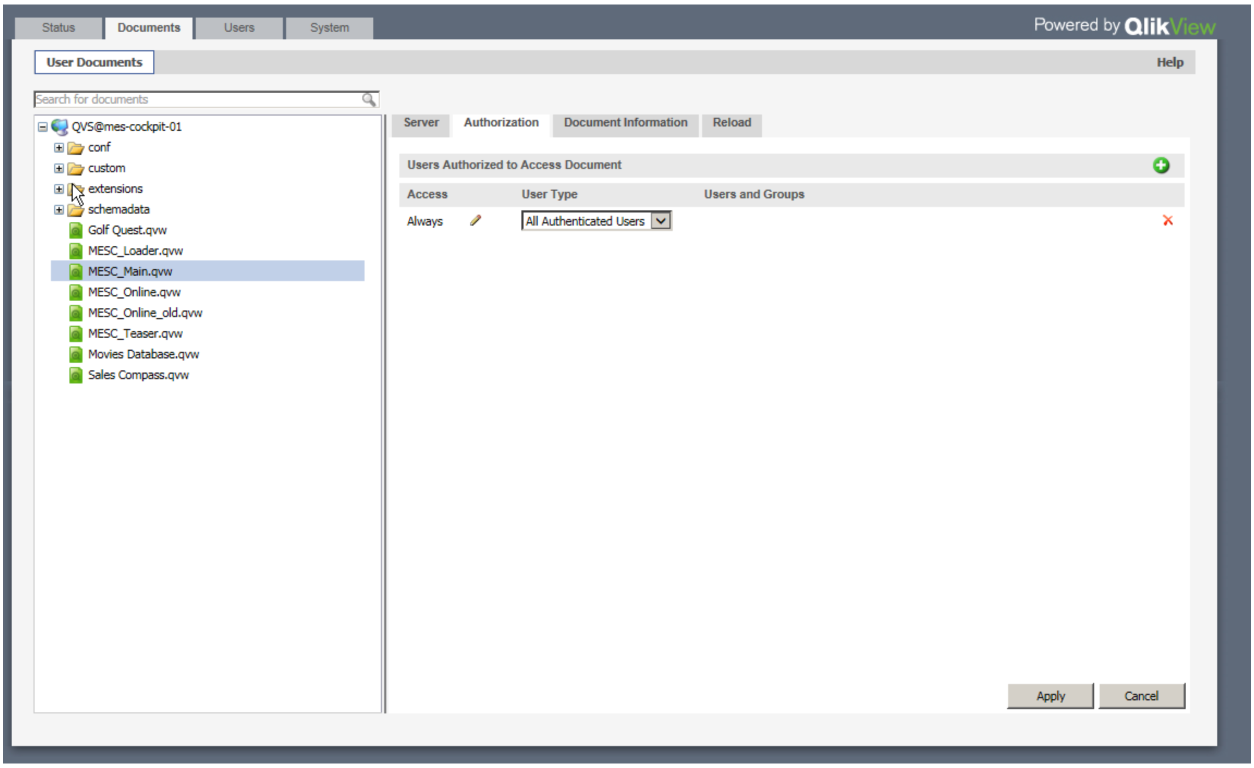
Production Monitoring: C:\ProgramData\QlikTech\Documents\MESC_Online.qvw

Performance Analysis: C:\ProgramData\QlikTech\Documents\MESC_Main.qvw

The original document must not be changed. Provided that modifications are required, the file has to be copied and renamed. Example: MESC_Online_<Customer's ID>.qvw

Only then changes to this customer-specific file can be made. The options provided in QlikView are described in the document: MESC_DevelopmentSuiteQV.pdf

If a new qvw file is copied or created, it must be accessible for authenticated users. To do so, the created file has to be assigned to the user type "All Authenticated Users" in Document --> Authorization in the management console of QlikView on the MES-Cockpit server.



1.2 Modifying Basic KPIs

Overview of benefits

Basic KPIs used for calculations and display in MES-Cockpit can be customized, i.e. further information may be added and/or they may also be restricted according to the customer's requirements.

Basic KPIs encompass all data exported from connected HYDRA systems and saved as *.xml file on the MES-Cockpit server for the calculation of KPIs. Customizing provides the following options:

- Additional data of already integrated Web Services for existing objects
- Integration of customer-specific basic KPIs for existing data types

By default, the following fields are exported:

Object "workplace":

MasterConfiguration:

Object	Data type	Web service	Fields
Workplace	Master data	BOResourceList	resource.id resource.designation resource.short_name resource.company

			resource.group resource.costcenter resource.blocked
	Online data	BOResourceList	"resource.id", "resource.short_name", "resource.group", "resource.costcenter", "resource.act.status.text", "resource.act.status", "resource.act.status.color", "resource.act.start_of_status", "resource.act.shift.total_time", "resource.cycle.target", "resource.cycle.actual", "resource.act.shift.yield.primary", "resource.act.shift.scrap.primary", "resource.act.shift.rpa1", "resource.act.shift.rpa2", "resource.act.shift.rpa3", "resource.act.shift.rpa4", "resource.act.shift.rpa5", "resource.act.shift.rpa6", "resource.act.shift.rpa7", "resource.act.shift.rpa8", "resource.act.shift.rpa9", "resource.act.shift.rpa10", "resource.act.shift.rpa11", "resource.act.shift.rpa12", "resource.act.strokes.total", "resource.act.strokes.yield"
	Key figure	Workplacebooking.list	"resource.id", "workplacebooking.shift.date", "workplacebooking.shift.number", "workplacebooking.rpa.number", "workplacebooking.yield.base", "workplacebooking.yield.primary", "workplacebooking.yield.secondary", "workplacebooking.yield.tertiary", "workplacebooking.scrap.base", "workplacebooking.scrap.primary", "workplacebooking.scrap.secondary", "workplacebooking.scrap.tertiary", "workplacebooking.rework.base", "workplacebooking.rework.primary", "workplacebooking.rework.secondary", "workplacebooking.rework.tertiary", "workplacebooking.problem.base", "workplacebooking.problem.primary", "workplacebooking.problem.secondary", "workplacebooking.problem.tertiary", "workplacebooking.strokes.total", "workplacebooking.status", "workplacebooking.status_duration", "workplacebooking.cycle.target", "workplacebooking.partitioning", "resource.pulse_factor.target" **<u>Calculated fields from MDC:</u>** w.status.class

			Total duration accrued for a status of the relevant status class w.oeee_arith *Performance is calculated on the MDE log record level. In this regard, all data records collected during the "production" status (RPA11) are used. Performance is calculated for each log record. For a compressed presentation in evaluations, these individual performance values are weighted. Weighting is based on production time (RPA11). The value exported for each workplace and shift represents the sum of weighted performance values.*
Operation	Master data	BOOperation.list	"order.id", "operation.id", "operation.costcenter", "operation.article", "operation.customerdesignation", "operation.articledesignation", "operation.ordertype", "operation.plan.yield.base", "operation.plan.yield.primary", "operation.plan.yield.secondary", "operation.plan.yield.tertiary", "operation.act.yield.base", "operation.act.yield.primary", "operation.act.yield.secondary", "operation.act.yield.tertiary", "operation.act.scrap.base", "operation.act.scrap.primary", "operation.act.scrap.secondary", "operation.act.scrap.tertiary", "operation.act.rework.base", "operation.act.rework.primary", "operation.act.rework.secondary", "operation.act.rework.tertiary", "operation.act.problem.base", "operation.act.problem.primary", "operation.act.problem.secondary", "operation.act.problem.tertiary", "operation.processing_time"
	Online data	BOOperationListCurrentlyLoggedOn	"operation.id", "operation.article", "operation.articledesignation", "order.id", "bookingrelation.logon_ts", "operation.designation", "operation.plan.yield.primary", "operation.act.yield.primary", "operation.act.scrap.primary", "bookingrelation.workplace", "bookingrelation.rpa1", "bookingrelation.rpa2", "bookingrelation.rpa3", "bookingrelation.rpa4", "bookingrelation.rpa5", "bookingrelation.rpa6",

			"bookingrelation.rpa7", "bookingrelation.rpa8", "bookingrelation.rpa9", "bookingrelation.rpa10", "bookingrelation.rpa11", "bookingrelation.rpa12", "operation.act.first_logon_ts", "operation.act.last_logoff_ts", "operation.act.last_interruption_ts", "operation.act.status", "operation.act.status.textnumber", "operation.plan.start_ts", "operation.plan.end_ts", "operation.earliest_start_ts", "operation.latest_end_ts", "operation.scheduled_start_ts", "operation.scheduled_end_ts", "operation.setup_time", "operation.processing_time"
	Key figure	DCAdeLogRecord.list	"operation.id", "orderbooking.shift.start_ts", "orderbooking.shift.number", "orderbooking.rpa1", "orderbooking.rpa2", "orderbooking.rpa3", "orderbooking.rpa4", "orderbooking.rpa5", "orderbooking.rpa6", "orderbooking.rpa7", "orderbooking.rpa8", "orderbooking.rpa9", "orderbooking.rpa10", "orderbooking.rpa11", "orderbooking.rpa12", "orderbooking.yield.base", "orderbooking.yield.primary", "orderbooking.yield.secondary", "orderbooking.yield.tertiary", "orderbooking.scrap.base", "orderbooking.scrap.primary", "orderbooking.scrap.secondary", "orderbooking.scrap.tertiary", "orderbooking.rework.base", "orderbooking.rework.primary", "orderbooking.rework.secondary", "orderbooking.rework.tertiary", "orderbooking.problem.base", "orderbooking.problem.primary", "orderbooking.problem.secondary", "orderbooking.problem.tertiary", "orderbooking.labor_utilization"
Order	Master data	BOOrderOverview	"order.id", "order.costobject", "order.article", "order.projectnumber", "order.customerdesignation", "order.salesorder", "order.articledesignation", "order.type",

			"order.act.status.text", "order.act.status.color", "order.last_op_logoff_ts_calc", "order.no_recordable_op", "order.no_finished_op", "order.act.rpa1", "order.act.rpa2", "order.act.rpa3", "order.act.rpa4", "order.act.rpa5", "order.act.rpa6", "order.act.rpa7", "order.act.rpa8", "order.act.rpa9", "order.act.rpa10", "order.act.rpa11", "order.act.rpa12", "order.act.yield.base", "order.act.yield.primary", "order.act.yield.secondary", "order.act.yield.tertiary", "order.act.scrap.base", "order.act.scrap.primary", "order.act.scrap.secondary", "order.act.scrap.tertiary", "order.act.rework.base", "order.act.rework.primary", "order.act.rework.secondary", "order.act.rework.tertiary", "order.act.problem.base", "order.act.problem.primary", "order.act.problem.secondary", "order.act.problem.tertiary", "order.plan.yield.base", "order.plan.yield.primary", order.plan.yield.secondary", "order.plan.yield.tertiary", "order.plan.lead_time", "order.plan.total_setup_time", "order.plan.processing_time", "order.plan.execution_time", "order.act.retention_period", "order.act.lead_time", "order.act.labor_utilization", "order.act.processing_time", "order.act.occupancy_time", "order.act.standstill_period", "order.act.setup_time", "order.act.wait_time", "order.first_op_logon_ts", "order.last_op_logoff_ts", "order.scheduled_start_ts", "order.scheduled_end_ts", "order.plan.labor_utilization", "order.ordertype"
	Online data	BOOrderOverview	"order.id", "order.article", "order.articledesignation", "order.salesorder",

			"order.act.status.led", "order.act.status.text", "order.act.status.color", "order.act.last_posting_ts", "order.earliest_start_ts", "order.scheduled_end_ts", "order.latest_end_ts", "order.scheduled_start_ts", "order.plan.yield.base", "order.act.yield.base", “order.act.scrap.base", "order.plan.lead_time", "order.plan.total_setup_time", "order.plan.processing_time", "order.plan.labor_utilization", "order.plan.execution_time", "order.act.retention_period", "order.act.lead_time", "order.act.setup_time", "order.act.processing_time", "order.act.standstill_period", "order.act.occupancy_time", "order.act.labor_utilization"
Operation_scrap	Key figure	DCAdeLogRecord.list	Scrap per scrap reason (orderbooking.scrap.reason)

1.2.1 Additional data of already integrated Web Services for existing objects

The following definition can be used to export further data for existing objects, such as for the "workplace" object:

By default, the single DataGetter functions define which data is exported via WebServices. This definition may be overwritten by customer-specific definitions in the config.xml file of MDC.

A template that may be changed is included in the MDC templates. MDC templates can be found here: `c:\ProgramData\mpdv\mdc\templates\`

The MDC templates include DataGetter functions for individual objects. Subject to the area from which data is to be taken, definitions have to be made in the xml file which also results in the data to be filed accordingly.

The following definitions can be made for existing objects:

Object	DataGetter	Data type	Web service	Config.xml
Workplace	WorkplaceDataGe	Master data	BOResourceList	MasterResultParameter

	tter	Online data	BOResourceList	OnlineDataResultParameter
		Key figure	Workplacebooking.list	KeyFigureResultParameter
Operation	OperationDataGetter	Master data	BOOperation.list	MasterResultParameter
		Online data	BOOperationListCurrentlyLoggedOn	OnlineDataResultParameter
		Key figure	DCAdeLogRecord.list	KeyFigureResultParameter
Operation_scrap	OperationDataGetter	Key figure	mescarchiveloadoperationscrap	KeyFigureResultParameter
Order	OrderDataGetter	Master data	BOOrderOverview	MasterResultParameter

All data from the "Keyfigure" area are exported as a total for each shift.

Customizations have to be inserted in the following MDC configuration file: `C:\ProgramData\mpdv\mdc\config.xml`

Example:

```
<ComponentConfiguration Name="WorkplaceDataGetter">

    <Parameter Name="MasterResultParameter"
    Value="resource.userfield02,resource.id,resource.designation,resource.short_name,
    resource.company,resource.group,resource.costcenter,resource.blocked"/>

    <Parameter Name="KeyFigureResultParameter" Value="<New field>,<List of
    key figures>"/>

    <Parameter Name="OnlineDataResultParameter" Value="<New field>,<List of
    online data>"/>

</ComponentConfiguration>
```

Please note: If a DataGetter function is integrated, only the fields listed in "Value" are requested by the WebService. This means: if a field is to be added all fields that should be exported for this object should be listed. In addition, every time data has been changed, the MDC service has to be restarted in MES-Cockpit server services.

1.2.3 Integration of customer-specific basic KPIs for existing data types

The defined XML file interface described in the customizing document entitled MESC_DevelopmentSuiteInterface.pdf can be used to integrate customer-specific basic KPIs.[1]

The requirements for using the XML file interface are:

- The structure described in the relevant document has been adhered to
- Data is transferred for existing data types
- Meta data/master data of objects are included
- Data to be transferred has to be filed in a new directory on the MES-Cockpit server. The directory name starts with the object name and an underscore character "_", e.g. Workplace_

The defined storage locations can be found in the configuration file MESC_DataLocations.txt (c:\ProgramData\QlikTech\Documents\conf\)

MESC_DataLocations.txt - Editor

```
BufferObj;BufferType;BufferDir
Responsibility;Master;$(vMasterDataPath)\responsibility
Responsibility;Master;$(vCustomDataPath)\responsibility
Formula;Master;$(vMasterDataPath)\formula
Workplace;Master;$(vMasterDataPath)\workplace
Workplace;Online;$(vOnlineDataPath)\workplace
Workplace;KeyFigure;$(vKeyDataPath)\workplace
Operation;Master;$(vMasterDataPath)\operation
Operation;Online;$(vOnlineDataPath)\operation
Operation;KeyFigure;$(vKeyDataPath)\operation\
Operation;KeyFigure;$(vKeyDataPath)\operation_scrap
Order;Master;$(vMasterDataPath)\order
Order;Online;$(vOnlineDataPath)\order
Translation;Master;$(vCustomDataPath)\translation
Translation;Master;conf\translation
User;Master;$(vMasterDataPath)\user
Standstill;Online;$(vOnlineDataPath)\standstill
TargetValue;Master;$(vMasterDataPath)\targetvalue
Schema;Schema;$(vSchemaDataPath)
```

1.3 Modifications to the Home Screen

Overview of benefits

The buttons displayed in the home screen of MES-Cockpit can be customized and/or additional buttons/icons may be added.

1.3.1 Customizing the first page of MES-Cockpit

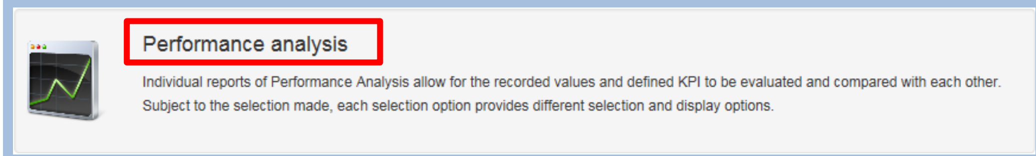
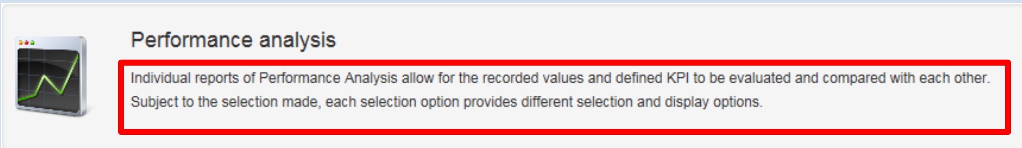
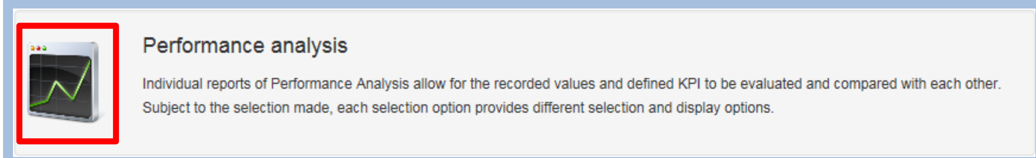
The file C:\inetpub\wwwroot\SMAMESC\Areas\Mesc\menu.xml includes the buttons/icons displayed in the home screen.

[1] Only customer-specific basic KPIs relating to HYDRA MES data may be integrated.

By default, the following buttons are shown in MES-Cockpit:

- Performance Analysis
- Production Monitoring
- Workplaces/ machines
- KPI monitor
- Contacts
- Messages listing

The following TAGs have to be defined in the menu.xml for each button:

TAG	Description
Label	LanguageKey displayed as caption in the button 
Description	LanguageKey of the brief description that is to be displayed in the button 
Icon	Path leading to the displayed icon of the button. The file has to be stored here: C:\inetpub\wwwroot\SMAMESC\Content\img 
Action	Function call to be started via the button.

Default sample configuration:

```
<?xml version="1.0"?>
<ArrayOfButtonView xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
```

```
<ButtonView>
<Label>lkMescPerformanceAnalysis</Label>
<Description>lkMescPerformanceAnalysisDescription</Description>
<Icon>/Content/img/Stats.png</Icon>
<Action> <Target>location='../mesc/qvcontainer?qvItem=MESC_Main'</Target>
</Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>

<ButtonView>
<Label>lkMescOnlineMonitor</Label>
<Description>lkMescOnlineMonitorDescription</Description>
<Icon>/Content/img/WebSystem.png</Icon>
<Action> <Target>location='../mesc/qvcontainer?qvItem=MESC_Online'</Target>
</Action> <DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>

<ButtonView>
<Label>lkWorkplaceOverview</Label>
<Description>lkWorkplaceOverviewSmaDescription</Description>
<Icon>/Content/img/Generators.png</Icon>
<Action> <Target>location='../resource/resource'</Target> </Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>

<ButtonView>
<Label>lkKeyMonitor</Label>
<Description>lkKeyMonitorSmaDescription</Description>
<Icon>/Content/img/ReportsPieChart.png</Icon>
<Action> <Target>location='../oee/kzm'</Target> </Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>

<ButtonView> <Label>lkContactPersons</Label>
<Description>lkContactPersonsDescription</Description>
<Icon>/Content/img/BusinessPartnersBlueMaleRedFemale.png</Icon> -<Action>
<Target>location='../ContactPersons/ContactPersons'</Target> </Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>
```

```
<ButtonView> <Label>lkMessageList</Label>
<Description>lkMessageListSmaDescription</Description>
<Icon>/Content/img/Generators.png</Icon> -<Action>
<Target>location='../MaintenanceManagement/MessageList'</Target> </Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>

</ArrayOfButtonView>
```

In addition to the configuration in the menu.xml file, the sub-folder "profiles" (C:\inetpub\wwwroot\SMAMESC\Areas\Mesc\Profiles) includes a configuration file for each integrated qvw file. This file includes the qvw file to be called up for the entry in the home screen according to the following pattern:

Example for MESC_Main.qvw:

```
<QvAppConfig xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">

<Server>http://swe-cbu-01</Server>
<Document>MESC_Main.qvw</Document>
<ApplicationName>lkMescPerformanceAnalysis</ApplicationName>
<ApplicationId>PerformanceAnalysis</ApplicationId>

</QvAppConfig>
```

1.3.2 Changes to existing buttons

If changes are required, they have to be made in a customer-specific menu_LOCAL.xml file. The entries from the original menu.xml file can be used as template. The file can be found here:

```
C:\inetpub\wwwroot\SMAMESC\Areas\Mesc\menu.xml
```

Available parameters and modification options are described in section 1.3.1.

1.3.3 Calling up a new evaluation/report with an existing button

If one of the existing QlikView files is to be changed, it has to be copied and renamed at first (e.g. MESC_Main_xxx.qvw). Then all changes/modifications can be carried out within the range of options provided by QlikView. The following measures have to be taken making sure that the new (changed) file is started and updated cyclically via the home screen instead of the previous QlikView file:

- Adjusting the relevant profiles file (C:\inetpub\wwwroot\SMAMESC\Areas\Mesc\Profiles) by defining the new document name, e.g. MESC_Main_xxx.qvw
  Example:

```
<QvAppConfig xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
<Server>http://swe-cbu-01</Server>
<Document> MESC_Main_xxx.qvw </Document>
<ApplicationName>lkMescPerformanceAnalysis</ApplicationName>
<ApplicationId>PerformanceAnalysis</ApplicationId>
</QvAppConfig>
```

- Changing the MDC and transferring the reload section to the config.xml file of MDC:

```
- <ComponentConfiguration Class="mpdv.MachineDataCollector.ConnectorPlugin.Qlik.QmsCallerComponent" Name="ReloadOnline">
    <Parameter Name="QmsServiceUri" Value="http://[SERVER]:4799/QMS/Service"/>
    <Parameter Name="TaskName" Value="Reload of MESC_Online.qvw"/>
    <Parameter Name="TaskPassword" Value="mpdv"/>
    <Parameter Name="EventFilter" Value="onlinedata"/>
  </ComponentConfiguration>
```

1.3.4 Integration and starting of a new, additional evaluation/report

When integrating a new qvw file, it can be created based on the existing application. Please proceed as follows:

By copying an existing QlikView file, a new one will be created, e.g. MESC_Main.qvw. Proceed as follows making sure this one will be displayed, called up and updated as an independent button/icon in the home screen in addition to the existing ones:

1. The new entry has to be added to the file menu_LOCAL.xml (C:\inetpub\wwwroot\SMAMESC\Areas\Mesc) displaying the buttons/icons. To do so, an already existing configuration for e.g. MESC_Main can be copied and adjusted accordingly.
   - o

```
<ButtonView>
<Label>lkxxx</Label>
<Description>lkxxxDescription</Description>
<Icon>/Content/img/xx.png</Icon>
<Action> <Target>location='../mesc/qvcontainer?qvItem=<Name of
the qvw-Datei withour Extension>'</Target> </Action>
<DoNotDissMissModal>false</DoNotDissMissModal>
</ButtonView>
```

Please note: The entered language key must be maintained in the relevant translation file and the defined icon must be created in the relevant path.
The name of the icon has to start with u_.

2. A new profile configuration file (C:\inetpub\wwwroot\SMAMESC\Areas\Mesc\Profiles) must also be created for the new qvw file. To do so, an existing configuration file can be copied, renamed and modified accordingly. Example:

```
<?xml version="1.0"?>
<QvAppConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
<Server>http://swe-cbu-01</Server>
<Document>MESC_Main_xxx.qvw</Document>
<ApplicationName>lkxxx</ApplicationName>
<ApplicationId>xxx</ApplicationId>
</QvAppConfig>
```

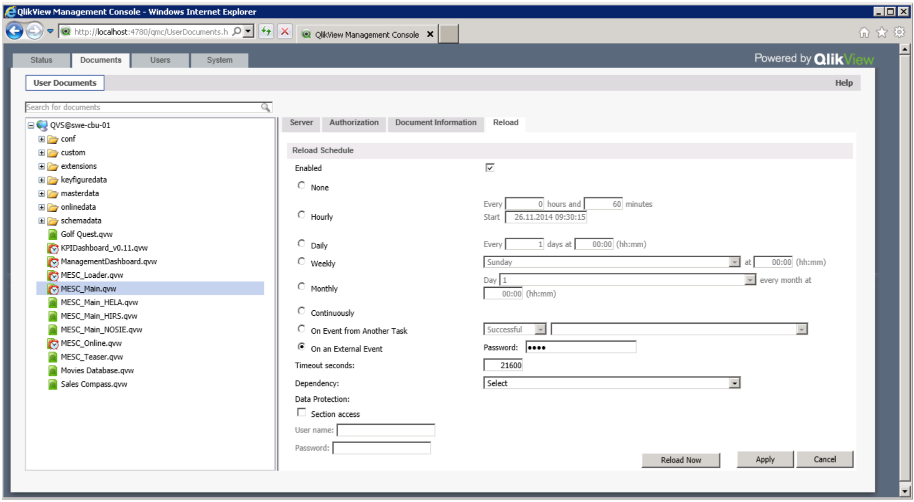
A new "ApplicationID" must be assigned.

3. Organize reload

The following alternatives can be configured in order to reload data and/or the qvw file including the data in cyclic intervals:

- Definitions via the QlikView Management console

  A reload may be defined for a stored document.



  Either for a cyclic event, e.g. once a day at 7.00 a.m. or it is also possible to attach the reload to an "external event". In this case the user's password must be entered and reloading is triggered by the MDC. In this case, the following step must also be carried out:

- Integrating the reload into the MDC

  Proceed as follows to trigger the reload process by MDC:

  - The following paragraph has to be entered in the config.xml file of MDC:

```
- <ComponentConfiguration Class="mpdv.MachineDataCollector.ConnectorPlugin.Qlik.QmsCallerComponent" Name="ReloadOnline">
    <Parameter Name="QmsServiceUri" Value="http://[SERVER]:4799/QMS/Service"/>
    <Parameter Name="TaskName" Value="Reload of MESC_Online.qvw"/>
    <Parameter Name="TaskPassword" Value="mpdv"/>
    <Parameter Name="EventFilter" Value="onlinedata"/>
  </ComponentConfiguration>
```

  - Please note:
    - A new name must be entered

- The name of the new qvw file has to be entered in "Value" of the parameter "task name"
- The event filter has to be set accordingly: "online" for online data or "masterdata,keyfiguredata" for master data and key figure data

Error! Reference source not found.

1.4 Further customizing options

1.4.1 Customer-specific translations

Overview of benefits

The displayed texts can be customized for the single areas.

1.4.1.1 Translations in Performance Analysis and Production Monitoring

All texts displayed in the applications can be renamed according to the customer's requirements by using a customized resource file.

To do so, the language keys (lk...) and relevant translations have to be entered in the Excel file MescDictionary.xlsm (.\conf\translation\MescDictionary.xlsm) and then the resource file has to be generated by clicking the "Export" button and filed here:

Default path: `C:\ProgramData\QlikTech\Documents\conf\translation`

Customized path: `C:\ProgramData\QlikTech\Documents\custom\translation`

Once the customer-specific resource file has been exported, the file .\custom\translation\translation.qvd has to be deleted.

The variable LOCALIZE can be used to access translations included in QV applications (example: $(LOCALIZE('lkYield'))). It replaces a LanguageKey by the translation defined in the user's language.

1.4.1.2 Translations in the First Page of MES-Cockpit

All texts of the start page have to be maintained in the "SMA" area and/or they may be renamed according to the customer's requirements in a customer-specific resource file.

To do so, the language keys (lk...) and relevant translations have to be entered in the Excel file mpdvDictionaryCustomer.xltm and then the resource file has to be generated by clicking the "Export" button and filed here:

```
C:\inetpub\wwwroot\SMAMESC\Runtime\resources\standard\languages
```

1.4.2 Definition of refresh rates

Data exports from connected HYDRA systems are triggered in cyclic intervals. Different triggers are used depending on whether online data or basic KPIs are exported. The following cycles are defined by default:

- Exporting status information: every 1.5 minutes (90 seconds)
- Exporting basic KPIs: every 30 minutes (1800 seconds)

The MDC attempts to trigger the export based on the above-mentioned cycles. Reasons why triggering fails and no data is exported:

- If data is currently being exported (long export times), the new export process will not be started
- A new export process can only be initiated, once reloading of data to the qvw file has been finished
- Reloading of all qvw files must be completed before starting a new reload process, i.e. by default the files MESC_Main and MESC_Online - there are dependencies between both qvw files.

If the cycle is to be changed, the relevant part from the template file (C:\ProgramData\mpdv\mdc\templates\config_Mesc.xml) including the modified value must be copied into the config.xml file of MDC. Example:

Basic KPIs "order"	<Parameter Name="[GetMasterDataOrder]" Value="1800" />
Target values	<Parameter Name="[GetMasterDataTargetValue]" Value="1800" />
Responsibility areas	<Parameter Name="[GetMasterDataResponsibility]" Value="1800" />
Formulas	<Parameter Name="[GetMasterDataFormula]" Value="1800" />
Master data "operation"	<Parameter Name="[GetMasterDataOperation]" Value="1800" />
Master data "workplace"	<Parameter Name="[GetMasterDataWorkplace]" Value="1800" />
Basic KPIs "operation"	<Parameter Name="[GetKeyFigureDataOperation]" Value="1800" />
Basic KPIs "OP scrap evaluation"	<Parameter Name="[GetKeyFigureDataOperation_Scrap]" Value="1800" />

Basic KPIs "workplace"	<Parameter Name="[GetKeyFigureDataWorkplace]" Value="1800" />
Online data "workplace"	<Parameter Name="[GetOnlineDataWorkplace]" Value="90" />
Online data "operation"	<Parameter Name="[GetOnlineDataOperation]" Value="90" />
Online data "order"	<Parameter Name="[GetOnlineDataOrder]" Value="90" />
Online data "downtime hit list"	<Parameter Name="[GetOnlineDataStandstill]" Value="90" />

If the section "TimedTrigger" is not included in the config.xml file, it has to be added. Please note in this context that the parameter "class" has to be removed.

1.4.3 Integration of new HYDRA systems

1.4.3.1 Integration of New HYDRA Systems into Performance Analysis and Production Monitoring

Connected systems from which data is exported can be found here:

```
c:\ProgrammData\mpdv\mdc\config.xml
```

For each connected HYDRA system there is an entry that is distinctly specified by the attribute "NumberOfLines" .

Each HYDRA system to be connected to MES-Cockpit must be entered in the configuration of MDC. The attribute "NumberOfLines" is to be set to the number of configured systems.

```
<ComponentConfiguration Name="WebServiceConnectorX">

    <Parameter Name="SystemName" Value="<Name>" />

    <Parameter Name="Host" Value="<Host>" />

    <Parameter Name="Port" Value="<Port>" />

    <Parameter Name="User" Value="<User>" />

    <Parameter Name="Pwd" Value="<Pwd>" />

</ComponentConfiguration>
```

A customer-specific identifier should be entered as the SystemName (maximum length 15 characters; [0-9a-zA-Z]). As an alternative, the host name can be entered.

Please note: If the configuration is changed and to enable these changes the MPDV MachineDataCollector service has to be restarted. Please also take into account that already exported data have actually been exported for the previously defined system. (Exported data include the name of the system they refer to. If this name is changed, please note that "old" data still include the "previous" name).

1.4.3.2 Integration of New HYDRA Systems in Production Information

All HYDRA systems to be evaluated in the Production Information area have to be entered in the following file:

```
…\SystemX\HyInstMgrDir\Instancerepo.properties
```

Example:

Configuration for the Hydra8 administration system "mesc-server:8080“ and the systems "hydra-server1:8080“, "hydra-server2:8082“:

```
=======================================
instance.name.1=MESC-ADMIN-SERVER

instance.name.2=HYDRA-SERVER
instance.name.3=HYDRA-SERVER2

MESC-ADMIN-SERVER.host.1=mesc-server
MESC-ADMIN-SERVER.port.1=8080
MESC-ADMIN-SERVER.id.1=1
MESC-ADMIN-SERVER.regserver.host.1=mesc-server
MESC-ADMIN-SERVER.regserver.port.1=6000
MESC-ADMIN-SERVER.description.1=MESC Admin

HYDRA-SERVER.port.1=8080
HYDRA-SERVER.host.1=hydra-server1
HYDRA-SERVER.id.1=1
HYDRA-SERVER.regserver.host.1=hydra-server1
HYDRA-SERVER.regserver.port.1=6000
HYDRA-SERVER.description.1=Hydra Germany

HYDRA-SERVER2.port.1=8081
HYDRA-SERVER2.host.1=hydra-server2
HYDRA-SERVER2.id.1=1
HYDRA-SERVER2.regserver.host.1=hydra-server2
HYDRA-SERVER2.regserver.port.1=6000
HYDRA-SERVER2.description.1=Hydra France
=======================================
```

1.5 Tips and tricks

1.5.1 Renaming of exported sites

When implementing and/or connecting HYDRA systems, a name is defined that will be used as site ID for exported data as of the first export. If another name should be displayed in MES-Cockpit this can be changed via the configuration file MESC_PlantMapping.txt.

This file can be found here:
`C:\ProgramData\QlikTech\Documents\conf\MESC_PlantMapping.txt`

This file maps the plant/site ID, i.e. the system name that is also exported and the newly defined name.

Example:

```
PlantId;PlantName
Plant1;Germany
Plant2;France
Plant3;China
```

1.5.2 Storage locations of values

Storage locations are configured in the following file in order for QlikView to "find" the exported data:

`C:\ProgramData\QlikTech\Documents\conf\MESC_DataLocations.txt`

Examples from the default file:

```
Operation;Master;$(vMasterDataPath)\operation

Operation;Online;$(vOnlineDataPath)\operation

Operation;KeyFigure;$(vKeyDataPath)\operation\
```

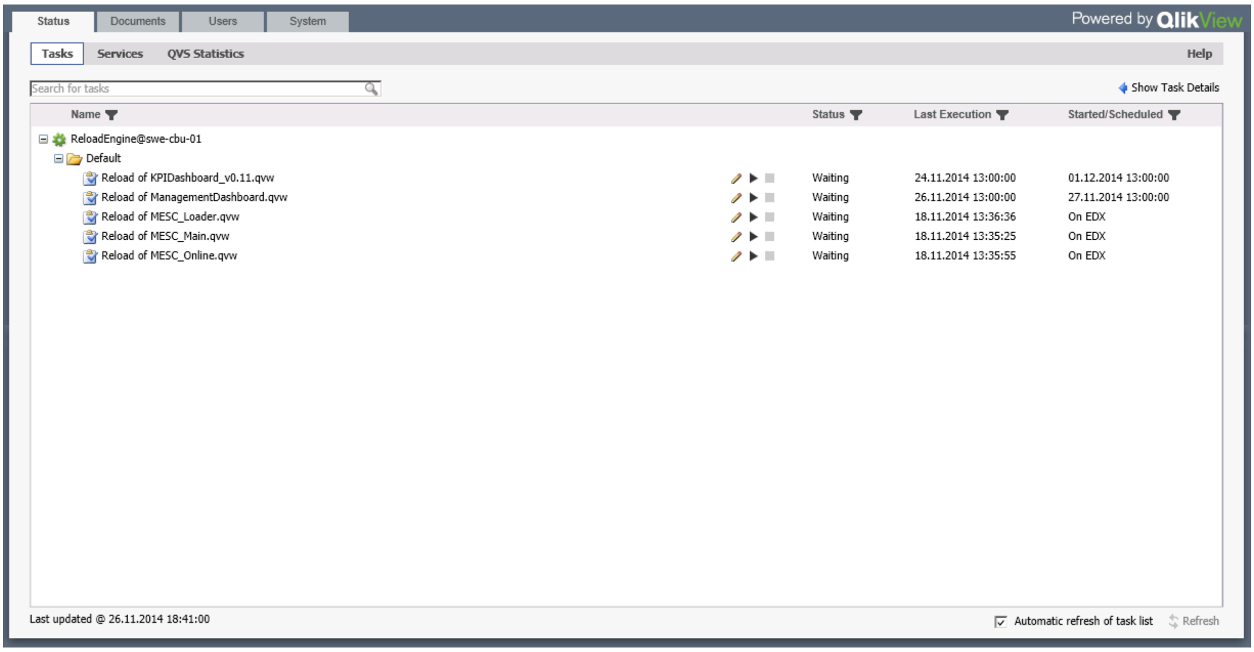
1.5.3 Defined variables used in QlikView

The following file includes the variables that can be used in and by QlikView:

`C:\ProgramData\QlikTech\Documents\conf\MESC_Variables.txt`

1.5.4 Last reload

The time stamp of the relevant QlikView file in the management console of QlikView shows the date and time the QlikView file was reloaded at last.


Status
Documents
Users
System
Powered by QlikView
Tasks
Services
QVS Statistics
Help
Search for tasks
Show Task Details
Name
Status
Last Execution
Started/Scheduled
ReloadEngine@swe-cbu-01
Default
Reload of KPIDashboard_v0.11.qvw Waiting 24.11.2014 13:00:00 01.12.2014 13:00:00
Reload of ManagementDashboard.qvw Waiting 26.11.2014 13:00:00 27.11.2014 13:00:00
Reload of MESC_Loader.qvw Waiting 18.11.2014 13:36:36 On EDX
Reload of MESC_Main.qvw Waiting 18.11.2014 13:35:25 On EDX
Reload of MESC_Online.qvw Waiting 18.11.2014 13:35:55 On EDX
Last updated @ 26.11.2014 18:41:00
Automatic refresh of task list
Refresh